central cavity (**spongocoel)** and goes out through the osculum. This water transport system is helpful in food gathering, circulation, respiration and removal of waste. **Choanocytes** or collar cells are special flagellated cells lining the spongocoel and the canals. The body is supported by a skeleton made up of calcareous and siliceous **spicules** or **spongin** or both. Nutrition is holozoic and intracellular. All sponges are hermaphrodites (i.e.) the ova and sperms are produced by the same individual. They also reproduce asexually by fragmentation or **gemmule formation** and sexually by the formation of gametes. Development is indirect with different types of larval stages such as **parenchymula and amphiblastula**.

Examples: *Sycon* (Scypha), *Spongilla* (fresh water sponge), *Euspongia* (bath sponge) *Euplectella* (Venus flower basket) (Figure 2.9).



Sycon



Hyalonema



Chalina



Euplectella

Figure 2.9 Examples of Porifera

The underwater sea bed is the new habitat where the discovery and development of Marine Pharmaceuticals are in peak. Anticancerous, Antimalarial drugs and other bioactive molecules have been isolated and tested successfully.

2.3.2 Phylum: Cnidaria

(G. *knode* -needle or sting cells)

Cnidarians (were previously called Coelenterata), are aquatic, sessile or free swimming, solitary or colonial forms with radial symmetry The name Cnidaria is derived from **cnidocytes** or **cnidoblasts** with **stinging cells** or **nematocyst** on tentacles. Cnidoblasts are used for anchorage, defense, and to capture the prey. Cnidarians are the first group of animals to exhibit tissue level organisation and are diploblastic. They have a central vascular cavity or **coelenteron** (serves both digestion and circulatory function) with a single opening called mouth or hypostome, which serves the process of ingestion and egestion. Digestion is both extracellular and intracellular. The nervous system is primitive and is formed of diffused nerve net. Cnidarians like corals have a skeleton made up of calcium carbonate. Cnidarians exhibit two basic body forms, polyp and medusa. The **polyp** forms are sessile and cylindrical (e.g. *Hydra, Adamsia*), whereas the **medusa** are umbrella shaped and free swimming. Cnidarians which exist



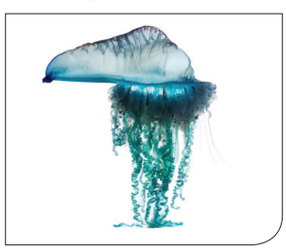
Adamsia



Pennatula



Meandrina



Physalia

Figure 2.10 Examples of Cnidarians

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